

Assignment No: – 6

ON

Full Stack Development – SSCA4020

AIM: Create a Simple Portfolio website Using React.

BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)

SUBMITTED BY:

Name: Shah Rishabh Alpeshbhai

Enrolment: 23SS02IT181

Class: BSCIT-H – 7A - Batch 2023-27

Commands

```
npm init -y
npm install react react-dom esbuild
npx esbuild src/index.jsx --bundle --outdir=dist --minify
```

src/index.jsx

```
import React, { useState, useEffect } from "react";
import { createRoot } from "react-dom/client";
import "./app.css";

/* ----- Component with props ----- */
function Header({ title, student }) {
  return (
    <header className="hdr">
      <h1>{title}</h1>
      <p>{student}</p>
    </header>
  );
}

/* ----- useState : Counter ----- */
function Counter() {
  const [count, setCount] = useState(0);
  return (
    <div className="card">
      <h2>Counter</h2>
      <div className="count">{count}</div>
      <button onClick={() => setCount(count + 1)}>+ Increment</button>
      <button onClick={() => setCount(count - 1)}>- Decrement</button>
      <button className="gray" onClick={() => setCount(0)}>Reset</button>
    </div>
  );
}

/* ----- Controlled input + live preview ----- */
function Greeting() {
  const [name, setName] = useState("");
  return (
    <div className="card">
      <h2>Controlled Input</h2>
      <input
        placeholder="Type your name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <p className="hello">{name ? `Hello, ${name}! Welcome to React.` : "Please enter your name"}</p>
    </div>
  );
}

/* ----- List rendering, events, conditional rendering ----- */
function TodoApp() {
  const [text, setText] = useState("");
  const [todos, setTodos] = useState([
    { id: 1, title: "Learn JSX", done: true },
    { id: 2, title: "Understand useState and useEffect", done: false },
  ]);

  const addTodo = () => {
    if (text.trim() === "") return;
    setTodos([...todos, { id: Date.now(), title: text, done: false }]);
    setText("");
  };

  const toggle = (id) =>
    setTodos(todos.map((t) => (t.id === id ? { ...t, done: !t.done } : t)));
  const remove = (id) => setTodos(todos.filter((t) => t.id !== id));

  // useEffect : runs whenever todos change
  useEffect(() => {
    document.title = `Todos (${todos.filter((t) => !t.done).length} pending)`;
  }, [todos]);

  return (
```

```

<div className="card wide">
  <h2>Todo List</h2>
  <div className="row">
    <input
      placeholder="Add a new task"
      value={text}
      onChange={(e) => setText(e.target.value)}
    />
    <button onClick={addTodo}>Add</button>
  </div>
  <ul>
    {todos.map((t) => (
      <li key={t.id}>
        <span className={t.done ? "done" : ""} onClick={() => toggle(t.id)}>
          {t.done ? "✓" : "○"} {t.title}
        </span>
        <button className="del" onClick={() => remove(t.id)}>Delete</button>
      </li>
    ))}
  </ul>
  {todos.length === 0 && <p>No tasks left!</p>}
  <p className="small">Pending: {todos.filter((t) => !t.done).length}</p>
</div>
);
}

function App() {
  return (
    <div>
      <Header title="React Interactive Web Page" student="Shah Rishabh Alpeshbhai | 23SS02IT181" />
      <div className="grid">
        <Counter />
        <Greeting />
        <TodoApp />
      </div>
    </div>
  );
}

createRoot(document.getElementById("root")).render(<App />);

```

src/app.css

```

body { margin: 0; font-family: Arial, sans-serif; background: #f0f4ff; }
.hdr { background: #20232a; color: #61dafb; text-align: center; padding: 18px; }
.hdr p { color: #ddd; margin: 4px 0 0; }
.grid { display: flex; flex-wrap: wrap; gap: 24px; justify-content: center; padding: 30px; }
.card { background: #fff; width: 280px; padding: 20px; border-radius: 12px; box-shadow: 0 4px 12px
  rgba(0,0,0,.12); text-align: center; }
.card.wide { width: 440px; text-align: left; }
.card h2 { margin-top: 0; color: #20232a; }
.count { font-size: 56px; font-weight: bold; color: #0b7285; margin: 10px 0; }
button { background: #0b7285; color: #fff; border: 0; padding: 8px 14px; border-radius: 6px; cursor:
  pointer; margin: 3px; }
button.gray { background: #868e96; } button.del { background: #e03131; padding: 4px 10px; font-size:
  12px; }
input { padding: 8px; border: 1px solid #999; border-radius: 6px; width: 70%; }
.row { display: flex; gap: 6px; } .row input { flex: 1; }
.hello { color: #2b8a3e; font-weight: bold; }
ul { list-style: none; padding: 0; } li { display: flex; justify-content: space-between; align-items: center;
  padding: 8px 0; border-bottom: 1px solid #eee; }
li span { cursor: pointer; } .done { text-decoration: line-through; color: #888; }
.small { color: #666; font-size: 13px; }

```

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>React App</title>
  <link rel="stylesheet" href="dist/index.css">
</head>
<body>
  <div id="root"></div>

```

```
<script src="dist/index.js"></script>
</body>
</html>
```

OUTPUT:

